10 The Respiratory System

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1. INTRODUCTION

Often overlooked in psychophysiology, the respiratory system is remarkably complicated and sensitive to a variety of psychological variables. Scientists have examined this system since the time of Galen and even before (Sternbach et al., 2001).

Over the intervening years some have argued for what has perhaps been an overly influential role for respiration. Feleky (1916) argued that the ratio between inspiration and expiration was intimately connected to psychological state and personality. Nielsen and Roth (1929) reviewed more than 20,000 spirometry records and identified ten breathing patterns that they argued were associated with the heritability of some traits. Sutherland, Wolf, and Kennedy (1938) stated the "respiratory curve is as constant and characteristic of an individual as his handwriting" and suggest that it is closely related to personality. Although few recent studies have sought to connect personality and respiration, it is clear that some individuals possess common respiratory patterns. This has led some to establish categories for these respiratory types or the "*personalite' ventalitoire*" (Shea & Guz, 1992).

More often than overestimating respiration's role in psychophysiology, this process is often ignored. In psychophysiology, respiration has been far overshadowed by research on the heart, brain, and electrodermal activity (Harver & Lorig, 2000). Wientjes and Grossman (1998) have effectively argued that it is respiration's susceptibility to voluntary influences that led early psychophysiologists to look elsewhere for physiological phenomena to study.

When respiration has been recorded, it is most often done to account for artifacts in other measures of more interest to the investigator (Grossman, 1983). Because of the coupling between breathing and cardiac output, heart rate changes as a function of the respiratory cycle. This phenomenon is called respiratory sinus arrhythmia (RSA) and for those interested in the cardiovascular system, it makes respiration an intervening variable. In fact, the degree of dissociation between heart rate and respiration has become of great interest in recent years since the "tightness" of this coupling can be viewed as an index of the vagal control of the heart (Berntson, Cacioppo, & Quigley, 1993). In addition to influencing cardiac activity, respiration can be a potent source of noise in electrodermal recordings as well (Rittweger, Lambertz, & Langhorst, 1997).

Part of the problem associated with measuring respiration in psychophysiology is that our needs are often so different from those of respiratory medicine. The past decade has provided tremendous advances in data and approaches related to respiratory health. Medical approaches to respiration are direct. They have been systematic and a number of standards exist for making lung function measurements (Evans & Scanlon, 2003). Almost all measurements of lung function in medical settings are made using spirometry. Even though this can be a very valuable technique in psychophysiology, many research questions limit the use of this approach. During a spirometric evaluation, the patient is usually nose clipped and breathes through a tube placed into the mouth. Though not invasive in the typically sense, it is psychologically invasive because it often centers the patient or subject's attention on the process of respiration (Han et al., 1997). In a medical setting, a technician making the recording encourages the patient to exhale even after the person feels their lungs fully exhausted. Certainly, this type of subject-experimenter interaction is foreign to most psychophysiological laboratories. Research questions seeking to relate cognition or emotional change to respiration must take a somewhat different approach and often use continuous measurements such as thoracic distention, pressure, or temperature as dependent variables while the tasks of interest are performed. These measurements provide less precise information about respiratory volume than spirometric approaches and don't often provide information about blood gasses. Even so, they may be preferred in some situations because they do not invade the psychological dimensions of the task. Carefully habituating subjects to spirographic evaluation can also be accomplished (Ritz, 2004) but requires great care and tasks that demand attention from subjects.

Few would argue that respiration's primary function is gas exchange. Oxygen is delivered to the blood stream

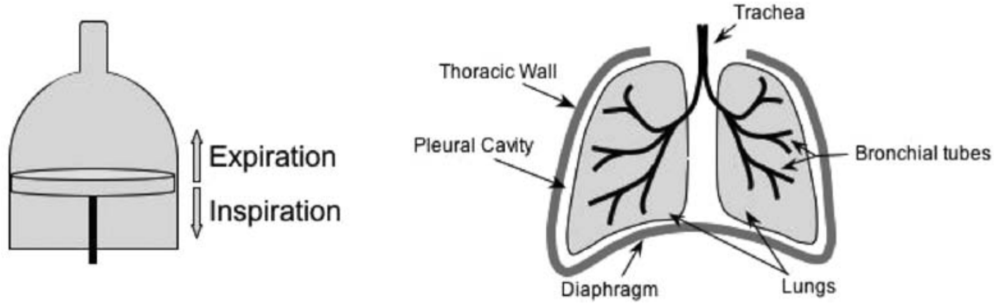

Figure 10.1. The left panel illustrates the principle by which the lungs operate. As the piston or diaphragm moves down, the pressure inside the vessel is reduced and air flows in to equalize the pressure. The right panel illustrates the placement of the lungs and diaphram in the thoracic cavity.

and carbon dioxide is removed by the actions of this system. Certainly the literature in medicine has emphasized this role and much research has been conducted to evaluate chemosensory receptors in the wall of blood vessels, absorption of oxygen into the blood, integrity of the lungs and alveoli, and many other portions of this system (Comroe, 1974). For psychophysiologists however, gas exchange is but one of several important functions. The respiratory system also acts to modulate the air necessary for speech, the pressure necessary to bring odors to the olfactory mucosa, and it anticipates the metabolic demands of cognitive and muscular activity (Collet et al., 1999). Furthermore, the interaction of this system with cardiac and other autonomic systems is a remarkable illustration of the functional complexity and cross-system integration of the nervous system.

Respiration is important in a number of illnesses of interest to psychologists such as asthma, which is widely studied for its psychological dimensions. From a respiratory perspective, it is associated with increased respiratory resistance or reduced airflow (Ritz, 2004). Persons experiencing an asthma attack must exert more effort to inhale because of narrowed air passages. This increased ventilatory effort is, in itself, a cue for breathing difficulty and may serve to provide positive feedback exacerbating respiratory difficulty (Harver & Mahler, 1998).

At its essence, the respiratory system is mechanical. Muscles control the filling of air sacks and these sacks are, in turn, connected to the outside world through air passages. Figure 10.1 illustrates the similarity of this system to a piston. The left panel of the figure shows a piston in a bottle. Movement of the piston is analogous to the contraction of the diaphragm. When the diaphragm contracts, the piston moves down, it creates an area of low pressure that causes air to flow into the bottle or lungs. The right panel of the figure illustrates the schematic nature of the mechanical system. Integrity of the air passages, sacks, and muscles all influence the system because it is controlled though multiple feedback loops. As an example, diminished air intake in one inhalation leads to subsequent increases in muscle activity producing increasing lung expansion and airflow. If resistance is increased in the system, volume will be decreased even though rate of flow is high in just the same way that a nozzle on water hose increases the speed of the water but decreases the volume. The mechanical nature of this system belies its complexity. Because respiration is an essential component of life, it broadly influences activity in other systems. Similarly, it is mutually dependent on many other systems in the body for its regulation.

2. NEURAL CONTROL

The fact that breathing "is a truly strange phenomena of life caught midway between the conscious and unconscious" (Richards, 1953) is only suggestive of the system's diverse sources of regulation. Clearly, breathing is under voluntary control and just as clearly, it is not. Under ideal conditions, respiratory activity is a harmonious product of both sources of control. This harmony may breakdown in cases of pathology or under periods of stress to the system.

2.1. Autonomic control of respiration

Like most muscles systems, respiration depends upon feedback loops for its regulation. This feedback arises from a large number of receptor types including mechanoreceptors and chemoreceptors. In the case of the respiratory system, receptors in the walls of the airways (slowly adapting receptors – SARs) increase their activity in response to stretch associated with the volume of inhaled air (Hlastala & Berger, 1996). SARs provide negative feedback to the system via the vagus nerve. The action of these receptors is responsible for the Hering-Breuer reflex, a lengthening of the exhalatory phase of the cycle in response to distention of the airway wall.

In addition to SARs, Rapidly Adapting Receptors also influence the system by way of the vagus. These receptors are found on the epithelium of the airways and are responsive to both mechanical and chemosensory irritation. Activity in these receptors results in reflexive coughing (Hlastala & Berger, 1996). Particulates in the air and some chemicals such as chlorine activate these receptors which are more dense in the upper airways. This system is distinct from the trigeminally based irritation sensors in the nasal cavity and mouth.

Feedback to the respiratory system is also influenced by chemoreceptors in the arterial system. Theses receptors are found in a variety of arteries but those most influential to regulation of the respiratory system are the carotid bodies that are found in the carotid arteries. A second and slightly less influential set of chemoreceptors, the aortic bodies, is found in the wall of the aorta. Both the carotid and aortic bodies communicate with the central nervous system by way of the glossopharyngeal nerve. Receptors in both structures are sensitive to circulating levels of blood gases including oxygen and carbon dioxide and also blood pH (Hlastala & Berger, 1996). Because of the extreme vascularity of the carotid bodies in particular, even small reductions in oxygen lead to increased activity in the receptors. Similarly, activity in the nerve is also increased when CO_2 is abundant. These changes in blood gas increase activity in the nerve and subsequently the muscles of respiration. Through this mechanism, hypoxic events lead to increased ventilatory effort often including prolonged inhalation and gasping (Fung & St. John, 1995).

The arterial system is just one source for chemosensory regulation of breathing. There are also centrally located receptors sensitive to oxygen, CO_2, and pH in cerebrospinal fluid. Research by Harada et al. (1985) has demonstrated that perfusion of a rat brain in vitro leads to with increased levels of CO_2 or decreased O_2 leads to increased phrenic nerve output. This nerve is responsible for the contraction of the diaphragm. Hlastala and Berger (1996) review these data and propose two distinct areas or chemoreceptive zones. Both zones are on the ventral surface of the medulla with the larger zone at the level of the VIIIth nerve and the other is more caudal and closer to the midline.

Central mechanisms coordinate the input from these mechano and chemoreceptors and generate output to control the muscle groups associated with breathing. Several different areas of the brain contribute to the automatic regulation of the system and these areas differ from those used in voluntary control. As an example of this differential regulation, individuals afflicted with Ondine-Curse syndrome selectively lose the ability to automatically regulate breathing but are able to breathe by voluntary effort. This syndrome can arise in patients who have undergone surgery to transect the ventrolateral cervical spinal cord (bilaterally). This surgery is sometimes used for regulation of intractable pain and severs ascending pain projections. Descending pathways to the phrenic nerve are also disturbed however. Persons undergoing this procedure must be artificially ventilated during sleep because conscious regulation of breathing is necessary for their survival.

The descending fibers associated with automatic respiration arise in three different areas of the brain stem (Hlastala & Berger, 1996). These three groups of nuclei are located bilaterally in the pons, ventral and dorsal medulla. All of these respiratory groups have cells that increase their responding during inhalation and provide excitatory output to muscles as well as cells that inhibit spinal areas associated with output to respiratory muscles systems. Pathways between the respiratory groups and in the spinal cord provide internal feedback and regulation.

2.2. Voluntary regulation

Voluntary regulation of breathing ultimately uses the same muscles as automatic regulation but the pathways to those muscles differ. For instance, the spinal pathways that ultimately lead to the phrenic nerve and the nerves that control the intercostals travel through the corticospinal tract in the posterior-lateral portion of the cord. Output to those same nerves and muscles associated with automatic control travel in ventral pathways in the cord. Additionally, the nuclei that give rise to the fibers in these pathways are quite different. Because most automatic control can be associated with the brain stem (including the pons), voluntary control is much more complicated and involves cortical and diencephalic regions (Hlastala & Berger, 1996).

Much of the feedback for voluntary regulation comes from somatic receptors that relay information about the relative distention of thoracic muscles. This input comes to somatosensory areas of the thalamus and is relayed to diverse parts of the cortex and basal ganglia (Bramann, 1995). Cortical areas interact with basal ganglia and the medullary respiratory groups in coordinating output to the nerves and muscles associated with respiration. Instigation of the voluntary act of breathing, like the instigation of other voluntary acts, obviously involves multiple cortical systems but remains largely unknown. Curiously, because breathing can produce artifactual findings in fMRI research, there is great interest in the cortical components of respiration. Recent research on sniffing has indicated different mechanisms associated with voluntary sniffs versus odors arising in nonvoluntary inspirations (Sobel et al., 1998). Findings in this domain are bound to add tremendously to our knowledge of the cortical components of voluntary respiratory activity.

3. THE MECHANICAL SYSTEM

As mentioned previously, breathing is clearly a mechanical activity that is mutually dependent on vast feedback networks. This mechanical system consists of several interconnected parts that all function in unison but can have relatively independent regulation. Nasal congestion, as an example, may take place without altering patency of airways closer to the lungs (Eccles, 1996).

3.1. Functional anatomy

3.1.1. Lungs

The lungs are often characterized as air filled sacks. In fact, they are collections of much smaller sacks called alveoli that are connected to air passages. These sacks are plentiful (Comroe (1974) estimated 30 million), richly vascularized, and thin-walled. The massive surface area (approximately

140 m^2 (Johnson & Miller, 1968)) of these sacks and their thin walls make gas exchange fast and efficient. There are two types of cells that make up the wall of the alveoli. Type I cells form most of the volume of this wall and are the epithelial cells through which gases are exchanged. Type II cells are less plentiful but produce a surfactant-like phospholipid that reduces surface tension across the wall of the alveolar sack leading to more rapid gas exchange. The bronchial tubes that serve the lungs form three distinct regions in the right lung each with its collection of millions of alveoli. These lead to formation of three lung lobes: the lower, middle, and upper. The left lung, which is slightly smaller than the right, has only two lobes. Volume of the normal adult human lungs is approximately 5l (Hlastala & Berger, 1996).

3.1.2. Muscles

The primary muscle of external respiration (breathing) is the diaphragm. This is a large domed muscle (see Figure 10.1) that separates the thoracic cavity from the abdominal cavity. When contracted, the muscle flattens increasing the volume of the thoracic cavity. This creates an area of low pressure inside the cavity and airflows into the lungs to fill the volume. The lungs, which are highly elastic, are passive in this process. Diaphragmatic muscle spasms, also known as hiccups, are common and not necessarily indicative of other pathology.

In addition to the diaphragm, muscles of the ribs also contribute to breathing. The intercostal muscles are interconnected to the ribs. The external intercostals are the most superficial and their contraction leads to the lateral expansion of the rib cage. The contraction of these muscles and the diaphragm produce the majority of the muscular effort to inhale a bolus of air (DeTroyer, Gorman, & Gandevia, 2003). Exhalation is largely passive and is the result of gravity and the elasticity of the rib fascia and thoracic cavity. Rapid relaxation of the external intercostals and diaphragm leads to rapid exhalations. The role of the internal intercostals in this operation is controversial. A variety of reports suggest that these muscles act in opposition to the external intercostals and pull the ribs down thereby assisting in forced exhalation. In experiments with dogs, De Troyer and colleagues (1985) found that the internal intercostals function to further expand the ribs. Thus these two sets of intercostals muscles tend to both overlap in function and both act to expand the rib cage. At the margins of the ribs, the shallow insertion angle of the muscles leads to slight competition between these muscle groups for “opening” the ribs. This suggests that whereas both diaphragm and intercostals work to expand the volume of the thorax during inhalations, exhalation and even forced exhalation, are largely passive. More recent findings in humans (Wilson et al., 2001) have confirmed the effects the internal intercostals on exhalations. They found that these muscles, functioning on the lower ribs, do contract to increase expiration. Forced expirations may also involve abdominal muscles that, through their contraction, force the diaphragm further up into the thorax pushing out additional air.

3.1.3. Airways

The trachea is the most commonly described portion of the airways but the nasal and oral cavities, larynx, and bronchial tubes are also important features of the airway. At the upper end of the airway, the nose is a remarkable air sampling system. Air is pulled through the two external openings (nares) into the nasal cavity. This cavity has three boney protuberances called turbinates that increase the turbulent airflow allowing a relatively small volume of air to reach the olfactory mucosa while the majority of airflow is pulled into the trachea. Despite the enormous volume of air that flows through this system, it is capable of resisting drying under most conditions. Muscles of the nares contract during respiration and can be used to evaluate breathing.

Breathing may also take place through the mouth in some cases. Unlike the nose, this system dries easily and is not typically the preferred route for inspiration despite having half the ventilatory resistance of the nasal channel. Expired air, which tends to have increased humidity due to gas exchange, is less problematic. Inhaled air passes into the pharynx and past the epiglottis. The epiglottis folds up during inhalation opening the airway and down during swallowing to prevent liquids from entering (Hlastala & Berger, 1996).

The next portion of the airway is the larynx. In humans, this cartilaginous enlargement of the airway includes the vocal chords and is responsible for speech. The thyroid gland also occupies space in this area. Enlargement of the thyroid or pathology of the larynx may affect the air passage and increase airway resistance.

The trachea begins immediately below the larynx and divides into the left and right bronchial tubes. Like the lungs, the right bronchus is slightly larger in diameter than the left. These bronchi divide again and again. After some 24 divisions or generations, they connect to the alveoli (Ritz et al., 2002).

The trachea and bronchial tubes are sections of U-shaped cartilage bound by muscle tissue lined with an inner layer of epithelial and support cells. These cells contain cilia and other cells that secret mucous. Inflammation of these epithelial cells is commonly known as *bronchitis* and leads to excessive mucous production. As the cilia move this mucous (and foreign objects) back toward the mouth, reflexive contraction of the internal intercostals and abdominals (coughing) pushes short bursts of air and mucous out of the mouth. Additionally, the swelling of this tissue increases ventilatory effort causing the muscles of the system to expend more energy and thus requiring more ventilation to address the energy demands of that increased load in a positive feedback loop. Obviously, this problem is exacerbated by exertion and helps explain the “exhaustion” often reported by persons recovering from bronchitis or other pulmonary obstructive diseases.

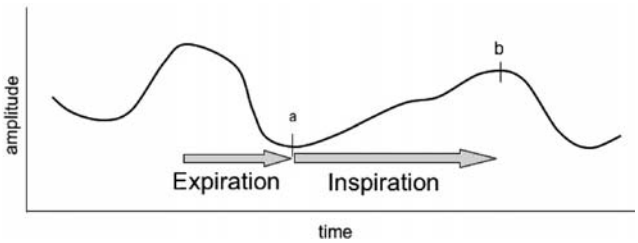

Figure 10.2. Typical output from piezoelectric belt transducer.

3.2. Measurement and quantification

Measurement and quantification of all types of pulmonary function have been aided immensely by the recent publication of guidelines for lung function measurements (Ritz et al., 2002). Readers should consult this report for specific details related to individual types of measurements. The descriptions of various techniques in this chapter are meant to help investigators make choices about which of the many respiratory parameters they may wish to evaluate.

Measurements of the mechanical aspects of respiration are straightforward and often quite simple. Most of these measurements depend upon changes in the volume of the thorax and abdomen and may be considered as a form of plethysmography. A variety of devices have been produced to make these measurements. Among the simplest is the respiratory belt (Grossman, 1967). This is a belt or tube that is affixed to the subject usually under the arms and around the chest and often a second belt is placed around the lower abdomen. The belt contains a sensor that is responsive to stretch. As the subject inhales, the volume of their upper thorax increases and stretches the device. In the past, these sensors have been, air filled bellows, strain gauges, mercury filled tubes, inductive coils and, most recently, fiber optic loops (Davis et al., 1999). The most commonly used belt method is currently a piezoelectric device. These are robust, don't leak air or mercury and provide reliable signals that can be incorporated into most electrophysiological recording systems. Pennock (1990) evaluated piezo-based respiratory belts and found that they were relatively linear compared to measurements made with a screen pneumotachometer. Figure 10.2 represents output typical of such a piezoelectric device.

The tracing in Figure 10.2 appears simple and is artifact free. Quantification of this signal is also straightforward but requires defining both the interests of the investigator and the parameters of the system. Peak exhalation is defined as the minima of the waveform between two cycles and is labeled as "a" on the graph. Peak inspiration is the maxima and is labeled as "b." It is an easy matter to evaluate the difference in these two points to find how much the chest changed during the inspiration. Counting the number of inspirations in a minute gives the rate of inspiration but the data contain far more information. Table 10.1 illustrates the variety of measurements that can be made from the simple waveform presented in Figure 10.2.

Only a few of these measurements are in common use but they serve to illustrate the complexity of the problem. Selection of the proper technique is related to the hypothesis, the way the data are acquired and the method used to reduce those data. Too often, an experimenter will seek to measure respiration then purchase the instrumentation to do so and find themselves at the end of the experiment left with still more questions about the respiration data. Perhaps worse, they will find themselves with a psychological variable that has no effect on *respiratory rate* and conclude that respiration was insensitive to their manipulation. In fact, many different parameters of respiration might have changed because *respiratory rate* is among the least sensitive (Wientjes & Grossman, 1998). This insensitivity is what led early researchers to adopt new metrics such as the inspiration/expiration ratio in the early twentieth century (Feleky, 1916).

In order to choose the proper technique, the investigator must consider, in depth, the mechanisms that lead physiological variables to change. If, for instance, one were concerned about cognitive problems related to subjects working at high altitude, the first mechanism to consider would be oxygen deprivation. Although such deprivation might lead to higher breathing rates, it will certainly lead to reduced blood levels of O_2 no matter the respiratory rate. In this example, the investigator should select a technique that, when mathematically reduced, will produce a direct measurement of the amount of oxygen in the blood over time so that it can be correlated with cognitive performance over time. It would be reasonable to add rate as a secondary measure but this would add relatively little to the findings. On the other hand, using respiratory rate exclusively would be highly problematic because rate is a very poor proxy for blood oxygen level. Respiratory patterns can be surprisingly complicated. A subject faced with a task may suspend breathing for several seconds. They may also immediately take a deep breath then suspend breathing. Breathing may become shallow or show a pattern of slow inhalations followed by brief periods of exhalations. Over minutes, one can expect that increases in metabolic demands will be equilibrated with both breathing and cardiovascular activity. Shorter epochs can be and often are much more variable and inconsistent with metabolic conditions. With respiration, just as with any psychophysiological dependent variable, the investigator must consider how the numbers that will ultimately be entered into their analysis will truly represent the psychological and physiological processes under investigation.

Although the respiratory belt has several advantages, it also suffers from several real problems. One problem is related to the transducer itself. Pulling the belt too tight leaves the transducer at the upper limit of its range and will produce "clipping" or a ceiling effect. Using a more loosely fitting belt can be equally problematic because only large inhalation may stretch the belt enough to produce a recording. Even with a properly fitting belt, individual differences in breathing lead some individuals to produce

Table 10.1. List of measurements that can be obtained from breathing transducers

Measurement	How obtained
Frequency/Rate	Mean number of peaks per minute
Inspiratory Volume (relative)	Base to peak or integration
Expiratory Volume (relative)	Base to peak or integration
Inspiration/Expiration Ratio	Computed
Inspiratory duration (absolute)	Base to peak duration
Expiratory duration (absolute)	Base to peak duration
Duration ratio	Computed
Complexity (frequency)	Spectral analysis combined with examination of the amplitudes of non-dominant frequencies

more abdominal distention making a belt placed under the arm pits insensitive to breathing. It is possible and often preferable to use a second belt placed over the abdomen (Grossman, 1967). Doing so, however, may increase subject discomfort. No mater how many belts are used, this method always provides external feedback because the belt must tighten during breathing resulting in greater constriction of the chest wall. Such feedback can make subjects far more aware of their breathing pattern and more likely to adopt response strategies that are different from less invasive procedures (Han et al., 1997). Finally, because the subject is often seated during testing, non-respiratory movements can stretch the transducer causing artifacts.

Similar to the respiratory belt is inductive plethysmography, here the subject wears a broad band and more recently, a "shirt" that contains sensors responsive to abdominal stretch and distention. The LifeShirt® (Wilhelm, Roth, & Sackner, 2003) is a new and interesting technique that may prove very helpful in the field of respiratory measurement. This is a continuous ambulatory monitoring garment for cardiovascular and respiratory measures (VivoMetrics, Ventura, CA) that collects a variety of data available for telemetry or later download and analysis. Like impedance pneumography, investigators attracted to using this device to obtain continuous cardiovascular measures will also get respiratory signals from inductance plethysomography recording devices. This technique uses changes in magnetic flux of wires embedded in the shirt to estimate thoracic volume and respiratory activity. It provides data on rib and abdominal excursions that lead to the calculation of many respiratory parameters including tidal volume, respiratory timing, and inspiratory flow. The system can also contain noninvasive blood gas monitors in addition to a variety of sensors not associated with respiration. As several reviews point out, techniques such as this offer the opportunity to make detailed measurements of subject physiology in real-life settings and emergencies.

Another technique that makes use of the distention of the chest wall and abdomen is magnetometry. In this technique, as the subject's abdomen moves, magnet flux is sensed and the position can be plotted. This 3-D space can be differentiated into a change score that can be viewed like the output from a respiratory belt. Levine and colleagues (Levine et al., 1991) report good results using sources placed on the ventral and dorsal surfaces in predicting tidal volume. Recently, McCool, Wang, and Ebi (2002) have introduced a portable version of this device.

In some situations such as a driving simulator or cockpit, it is possible to use cramped space to the experimenter's advantage. Casali, Wierwille, and Cordes (1983) used a proximity indicator to estimate the location of the abdomen during breathing. Although certainly not applicable to all situations, this technique had the advantage of being less invasive although it may be more susceptible to movement artifacts.

Whole body plethysmography refers to using the fluctuating volume of a chamber containing the subject to determine respiratory activity (Ries, 1989). The air supply to the subject originates outside the chamber. As air is pulled into the lungs, the change in chest volume reduces the volume of the chamber. In the past, the chambers were water filled but more recent approaches use an air filled chamber and pressure transducers. This remains a widely used method in medicine and has particular advantages for measuring lung capacity in laboratory animals (Rozanski & Hoffman, 1999).

Another important measurement of thoracic volume is impedance pneumography. This technique measures the impedance of the thorax as it changes during respiration. This is a viable and interesting technique but is not often employed. Even though few investigators actively seek to measure respiration this way, it is often used to measure the heart via impedance cardiography. Recently, Ernst and colleagues (Ernst et al., 1999) have devised a way to extract impedance pneumography data from impedance cardiography recordings. This approach uses standard tetrapolar electrodes for impedance cardiography. The pneumographic signal can be derived from ΔZ (or ΔZd derived by integration of dZ/dt) by first using a digital filter to eliminate high frequency and cardiac activity and then demeaning or removing the signal offset. Ernst and colleagues (1999) compared this derived signal with both spirometry and strain gauge recordings of the chest and found high coherences for the derived measure and spirometry. In fact, the derived signal was a

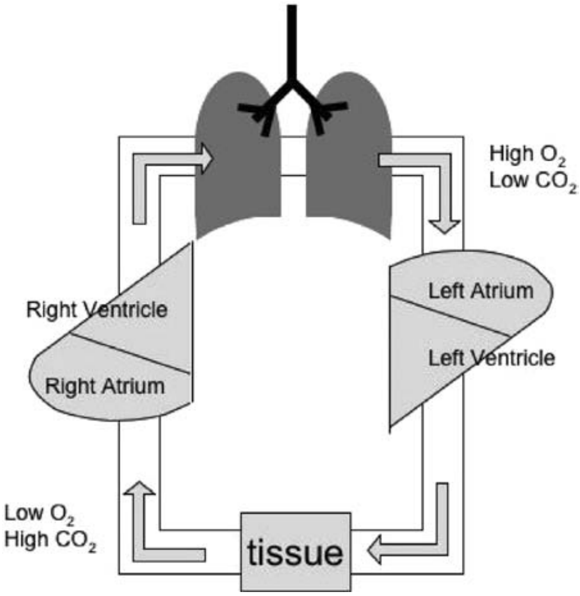

Figure 10.3. Function anatomy of gas distribution.

better predictor of the spirographic record than was the strain gauge. Because of the recent growth in impedance cardiography research, respiratory research may inadvertently gain new data and appreciation due to this analytic procedure.

Electromyography of the intercostals may also prove a useful technique under some conditions. If the investigator is measuring EKG from the chest wall in a position near the intercostals, these electrodes may also pick up activity from these respiratory muscles that, through filtering, can provide respiratory information (Pauly, 1957). Additionally, as mentioned previously, activity in the *nares dialator* can be monitored with electromyographic techniques (Sasaki & Mann, 1976).

4. THE GAS SYSTEM

The mechanical activity of the lungs, diaphragm and airways all serve the purpose of facilitating gas exchange with the blood. Oxygen in inhaled air passes through the wall of the alveoli and carbon dioxide diffuses from the blood into the alveoli and is expelled during exhalation.

4.1. Functional anatomy

Figure 10.3 shows a diagram of the functional anatomy of the gas distribution system. Air that is rich with oxygen and low in carbon dioxide fills the lungs and specifically, the alveoli. Here passive diffusion takes place across the cell membrane and pulmonary capillaries. Oxygen, in low concentration in the blood, flows into the capillaries and carbon dioxide that is in high concentration in the blood diffuses into the alveoli. This newly oxygenated blood flows into the left atrium and ventricle then provides the rest of the body a supply of oxygen. As cellular metabolism takes place, the amount of oxygen in the blood is reduced and the supply of carbon dioxide increased. These products flow through the blood into the right side of the heart and are pumped into the lungs where gas exchange occurs with the air (Guyton & Hall, 2000).

4.2. Measurement and quantification

There are many different approaches to measuring the gasses associated with respiration. Among the most simple and reliable is pressure (Carroll et al., 1992). Because inspired air flows into the lungs through the nose and mouth, a cannula connected to a pressure transducer can readily measure the pressure changes during breathing. Like the mechanical measurements mentioned above, the output of such a device produces a record very much like the respiratory belt. Unlike the belt, this type of measurement is far less prone to movement artifacts. Any movement of the cannula itself, however, can produce far less effective recordings. A variety of respiratory pressure devices are available that were primarily designed to monitor sleep apnea. Most use a small soft nasal cannula that fits in the nose and over the ears and is often used to deliver oxygen in home medical applications.

Similar to pressure transduction, temperature transducers may also be used to measure breathing. Because air is warmed in the lungs before being exhaled, a temperature sensor can readily detect exhalations. Inhalations can also cool the transducer if sufficiently forceful. Like the pressure transducer, this device is very sensitive to positioning and requires some means to fix its position on the face. Some transducers have a nose or mouth clip to facilitate this process. One additional caveat about this type of sensor is its response speed or time constant. Thermocouples and thermisters used in these devices tend to be rather slow and sluggish in response to rapid changes in temperature. A few devices of this type respond rapidly and investigators interested in the shape of the breathing cycle should be certain that they use a device fast enough to record the responses of interest. Although normal respiratory activity has a cycle of approximately 0.3 Hz (approximately 18 inspirations per minute), the shape of the wave will change much faster and require capturing phenomena of 2.0 Hz and possibly higher. As an example, inhalation of a malodor causes rapid cessation of the inspiration (Frank, Dulay, & Gestland, 2003). In cases such as this, a temperature transducer would be a very poor choice because it tends to be less sensitive to the inspiratory phase of the cycle and slow to react to the change in breathing.

Sound may also be an effective means of determining breathing cycle. A small microphone placed at the nares will record clearly different sounds for inhalation and exhalation allowing the investigator to make a determination of respiratory cycle. Such devices are most often used for recording snoring but can be adapted to record normal breathing sounds by integrating the output of the microphone. As one might imagine, this technique is prone to artifacts from ambient noise. Such artifacts can easily overwhelm the small changes that are due to respiratory activity. Furthermore, should the subject speak, the output from the microphone has the potential to saturate the amplifiers and delay subsequent recordings. Que and colleagues (Que et al., 2002) recently reported on a more sophisticated approach to this problem. Using a tracheal microphone, in a technique they call phonospirometery, they analyzed sounds related to air flow in the lungs and were able to estimate a variety of respiratory parameters including tidal volume. They report that their measures corresponded well to more classic procedures and did not require subject to use mouthpiece or nose clip.

Spirometry is the most widely used technique for evaluating respiratory activity and can be accomplished with a variety of devices. The original spirometers required subjects to use a mouthpiece and nose clamp. Their exhalations were used to displace water under a bell. The volume of the displaced water was used to determine the volume of gas (under pressure) that was expired. More sophisticated devices added simple gas analysis to this operation. Modern approaches to spirometry seldom use water displacement and gas analysis is now far more specific.

Although subjects continue to use a mouthpiece and nose clip, the air they exhale now flows through a tube and fills a bellows or activates a small turbine or pressure transducer attached to a screen. In fact, portable devices now make it possible to collect data with a handheld computer and lightweight spirometry tube that produce data that are comparable to laboratory spirometers (Mortimer et al., 2003). Spirometry tends to concentrate on one phase of the breathing cycle at a time and provides the experimenter with a volume X time plot of the inhalation or, more commonly, the exhalation. Almost all specific respiratory measurements are conducted using spirometery (see Table 10.1). These will be detailed later in this chapter.

In addition to measuring the volume of air inspired or exhaled, it is also common to assess the efficiency of the gas exchange using a gas analysis system. In such a system, the analytic apparatus samples expired air. This apparatus may be a general-purpose analytic system such as a gas chromatograph or a system designed to specifically analyze a particular respiratory gas such as carbon dioxide. In fact, carbon dioxide monitoring is relatively common. Both oxygen and carbon dioxide monitoring equipment provide numeric readouts and most provide a linear analog voltage referred to that readout for connection to recording equipment producing a volume X time recording. Investigators interested in this type of analysis should be concerned about the timing of the analysis. Some devices respond relatively slowly and may not give breath-by-breath responses making the volume X time recording less valuable.

Although not specifically related to spirometery or other respiratory measurements *per se*, non-invasive measurement of blood gas may be an important part of understanding respiration and its psychophysiological components. Recently, pulse oximetry has become available to record parameters related to blood gases. This technique requires that a transducer, much like a photoplethysmograph be placed on the subject's skin. The pulse oximeter provides an estimate of the percentage of hemoglobin that is saturated with oxygen. This device works similarly to a photoplethysmograph except that it detects the ratio of absorbed light for two different light sources, one of which is specifically absorbed by oxygen bound hemoglobin. This technique provides a rapid readout but is very sensitive to the nature of the vascular bed over which the sensor is placed (Allen, 2004).

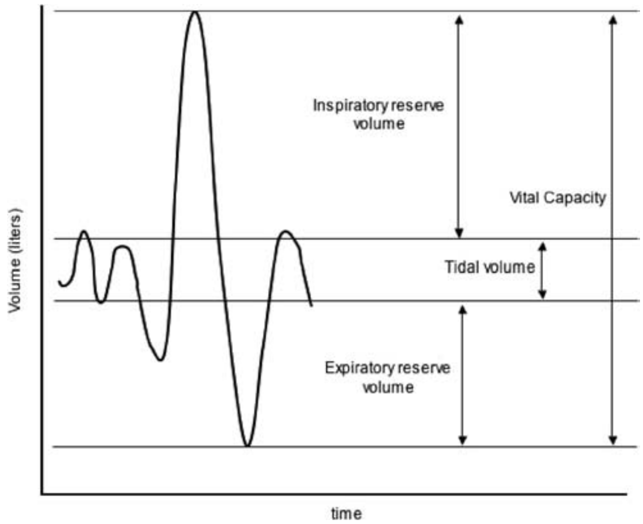

Figure 10.4. A typical spirographic record and the respiratory measurements that can be obtained from this method (adapted from Hlastala & Berger, 1996).

5. SPECIFIC RESPIRATORY MEASUREMENTS

Because of the importance of respiratory function to well-being and other physiological systems, it is extensively measured in medical settings. These measurements are standardized and can have a variety of uses for psychophysiologists interested in this system. Ritz and colleagues (2002), in their guidelines for psychophysiological research on respiration, provide a full list of such measurements. Furthermore, subjects undergoing this testing follow a prescribed protocol involving several forced breathing maneuvers. Typical values for many of these measurements are presented in Hlastala and Berger (1996) and some are discussed below. Figure 10.4 shows the relationship of several of these measures to a spirographic record. Some of the more commonly used measures are presented below.

Respiratory frequency (f) is the number of respiratory cycles occurring in one minute. This value ranges from about 12–15 respirations per minute under resting conditions.

Tidal volume (V_T) is the normal volume of air inhaled after an exhalation. It is highly variable but averages about 0.5l.

Vital capacity (V_C) is the volume of a full expiration. This metric depends upon size of the lungs, elasticity, integrity of the airways, and other parameters therefore it is highly variable between subjects. Values may range from 2l to 7l for adults.

Residual volume (V_R) is the volume that remains in the lungs following maximum exhalation. Like other measures that depend upon the size of the chest and lungs, this tends to be highly variable ranging from 1.41 to 1.91 in larger subjects.

Compliance (C) is a measure of the elasticity of the thoracic area. It may be applied to the lungs (C_L), chest wall (C_{CW}) or thorax (C_T). Conceptually it is the resistance to change in response to pressure. Highly compliant lungs, for instance, change in accord with external pressure changes. Thus, one may calculate compliance by determining the ratio of the change in volume (of the lungs or thorax) to changes in pressure. It should be noted that because this is an absolute measure, it is non-linear with respect to lung volume. Volume decreases disproportionately as lungs reach their maximal distention (Hlastala & Berger, 1996).

6. PSYCHOLOGICAL/FUNCTIONAL RELEVANCE

Because adequate ventilation is necessary for all physiological systems to function, it is little surprise that alteration of respiration leads to a cascade of changes throughout the body. Such changes affect the entire functioning of the organism including producing physiological and psychological effects. Some of these effects lead to artifacts in recording of the heart or functional imaging. But, as the preceding sections have shown, a diffuse network controls the respiratory system. The actions of this diffuse network lead many different psychological functions to alter respiration.

6.1. Respiratory effects as artifacts

6.1.1. Skin conductance

Skin conductance and other forms of electrodermal activity (EDA) are dramatically influenced by respiration. Asking subjects to make a deep breath is often used to validate the responsiveness and integrity of the recording system. Despite the sensitivity of EDA to respiratory changes, removal of artifacts due to respiration is difficult. EDA is multiply caused (Dawson, 2002) and rhythmic variations are not tightly coupled to respiratory cycle (Rittweger, Lambertz, & Langhorst, 1997). Recently, Schneider and colleagues (2003) have introduced a technique for identification of respiratory artifacts. It is too soon to know if this technique will enter widespread use for the correction of respiratory effects on EDA.

6.1.2. Heart

The effects of respiratory cycle on the heart are well known to psychophysiologists. The heart period, the time between successive beats, grows longer during exhalations leading to fewer beats per minute. During inhalation, heart period is shorter and consequently, heart rate appears to increase during this phase of respiration. This oscillatory interaction between the heart and respiratory system is know as respiratory sinus arrhythmia (RSA) and is the result of the influence of a variety of different physiological systems. Normally, evaluation of this interaction can produce an estimate of vagal tone in a subject. A full description of these interactions is beyond the scope of this chapter but the reader is referred to the chapter on cardiovascular psychophysiology in this volume for more extensive explanation of this phenomenon. Voluntary ventilation may introduce noise into this system and adversely influence estimates of vagal tone (Wilhelm, Grossman, & Coyle, 2004). Previous findings have indicated that even small changes such as missing an R spike can adversely affect such findings (Berntson & Stowell, 1998).

6.1.3. fMRI

Because functional magnetic resonance imaging (fMRI) is blood oxygenation level dependent (BOLD), it is reasonable to assume that the system that brings oxygen to the blood would be important in regulating the measured response. Consider this example. Subjects are asked to view a variety of pictures that are meant to evoke fear as well as a group of control pictures. Based on subjective ratings, the fear-producing pictures are effective in their emotional evocation and there are clear differences between the BOLD responses for the two picture types. One might conclude that the brain areas showing increased BOLD response during the fear pictures would be associated with the production of that emotion. In some ways they would, but not perhaps in the ways expected by the experimenters. Fear leads to increased ventilation rate and ventilatory effort (Boiten, Frijda, & Wientjes, 1994) due to airway constriction. Likewise, heart rate increases as does stroke volume (Sinha, Lovallo, & Parsons, 1992). Are the differences found in the brain's BOLD response the cause of these changes or their result? Furthermore, because of increases in stroke volume of the heart or the volume of the rib cage, the physical movement of the brain or volumetric disturbances in the magnetic field may impair the ability to accurately image the subjects during the fear evocation.

Correction of breathing and cardiac-related artifacts in fMRI is still controversial and a variety of papers have recently appeared suggesting several approaches. One of the most common is to gate or mask the responses during the different portions of the breathing cycle. Investigators might choose just the peak inspiratory phase of the cycle to make their measurements in BOLD responses or use a digital filter to isolate only BOLD changes in frequencies higher than respirations (Brosch, et al., 2002; Zhu et al., 2003).

6.2. Respiratory changes as a result of psychological influence

6.2.1. Emotion

Emotion has long been known to influence respiration. Descriptions of "breath taking" beauty and fear inducing an inability to "catch one's breath" have been a part of literature and folklore into the ancient past. As mentioned previously, Feleky (1916) recorded the ratio between inspiration and expiration and was able to identify a set of primary emotions. Others followed (Dudley, 1969), but relatively little research has been conducted related to emotion where respiration was the primary physiological variable of interest. Boiten, Fijda, and Wientjes (1994) review this literature and conclude that the major dimensions of emotion that alter respiratory activity are continua of calm-excited and active versus passive coping. They argue that it may be possible to make finer discriminations but the relative paucity of research and interest in this field has impaired our understanding of these phenomena. Subsequent research by Boiten (1998) has specifically addressed the effects of emotion on respiration using both chest and abdominal respiratory strain gauges. Subjects watched short video presentations designed to evoke a variety of emotions. Subjects altered their respiration patterns depending upon the content of the movies with more positive affect being associated with shorter inspiratory cycles. A scene that evoked disgust, led to respiratory pauses. Interestingly, without careful analysis, such pauses might be misinterpreted because long pauses would lead to lower respiratory rates. Such findings serve to illustrate the insufficiency of rate as a helpful measure of respiratory activity. Recently, Gomez, Stahel, and Danuser (2004) conducted a study evaluating respiratory responses to affective pictures confirming the importance of the continua proposed by Boiten and colleagues.

6.2.2. Asthma

Within psychophysiology, asthma is the most widely studied topic related to respiration. The National Institutes of Health report that, in the United States, approximately 15 million persons suffer from this chronic inflammatory disease of the airways. Asthma can be triggered by a reaction to environmental irritants but it is also often triggered by even the suggestion of irritants. Furthermore, it is clearly exacerbated by other stressors in sufferer's psychological environment. In their recent review of the psychological aspects of asthma, Lehrer et al. (2002) describe the nature of these stressors along with potential triggering mechanisms. They advocate the efficacy of symptom recognition and biofeedback for control of the cascade leading to airway constriction.

Ritz (2004) provides an excellent overview of this complicated and important area of research that is beyond the scope of this chapter. Just as the lungs are multiply controlled, so are the airways. The feedback, neural, and humoral regulation all interact and make delineating mechanisms of influence in this system difficult. Even so, the importance of this topic requires addressing the mechanisms at work.

6.2.3. Cognitive activity

Clearly, the reduction of the amount of oxygen in the blood stream can produce grave consequences related to brain function and cognitive ability. These hypoxic events may be related to extreme systems break down such as heart fibrillation, drowning, or suffocation but they may also be related to far less dramatic causes. Sleep apnea (Aloia et al., 2004), nocturnal asthma (Bender & Annett, 1999), and even scuba diving (Slosman et al., 2004) have some cognitive sequelae.

There is little evidence that breathing pattern, so long as it supplies sufficient oxygen, affects cognitive activity. The exception to this statement is a maneuver called "forced unilateral breathing." This exercise originated in yogic training and several studies have demonstrated that occlusion of the left or right nostril alters cognitive function. This is an interesting, varied and surprisingly large literature. Werntz, Bickford, and Shannahoff-Khalsa (1987) found that occlusion of the left or right nostril produces increased EEG amplitudes in the contralateral hemisphere. Contrary to most interpretations of increased EEG amplitude, this finding was interpreted as increased cortical activity and was used to account for behavioral differences associated with the "activated" hemisphere. A different study (Block, Arnott, Quigley, & Lynch, 1989), examined this breathing style on verbal test performance and spatial tasks finding effects for both gender, nostril and task. Males in this study performed spatial tasks better when breathing through the right nostril and the opposite response was found for women. A slightly more recent study by Jella and Shannahoff-Khalsa (1993) found that both men and women showed improved performance during left nostril breathing contradicting the earlier effect in males. Although inconsistent, the preponderance of the literature suggests that right nostril breathing is associated with better verbal performance and left nostril breathing is associated with better spatial performance. These findings have been interpreted as being the result of increased activation in the hemisphere contralateral to the nostril with airflow. Recently, this idea has been extended to research on emotional tones in a dichotic listening paradigm (Saucier et al., 2004).

Although a full review of this literature is inappropriate for this chapter, these studies rarely propose a mechanism by which uninostril breathing might alter hemispheric and/or cognitive activity. Nasal cycle, an alternating ultradian rhythm associated with increased patency of the left or right nostril, is poorly understood and often uncontrolled in these studies. Furthermore, the odor environment in which these studies are conducted is not described. Lorig et al. (1989) demonstrated that nasal versus mouth breathing had a profound influence on EEG and Lorig and coworkers (Lorig, 1999; Lorig, Malin, & Horwitz, 2005) later found that performance of verbal and other symbolic tasks can be impaired when the subject is challenged with an odor. Because the olfactory system projects ipsilaterally, left nostril breathing of odorized air could lead to poorer performance on verbal or symbolic tasks than right nostril breathing due to intrahemispheric competition. Because no air supply is truly odor-free, this simple and overlooked effect has the potential to account for much of the data in this literature. No study has, however, directly evaluated the connection of odor to the cognitive literature related to unilateral nostril breathing.

7. OTHER RESPIRATORY FUNCTIONS

Although breathing is normally associated with gas exchange, it is also an avenue for other functions related to the movement of air. As indicated in the preceding paragraph, odor acquisition is also a function of the respiratory system. Similarly, the muscles of respiration are also used for controlling the air for speech. Although such functions are certainly not the most important aspects of respiration, they are inextricably linked to the gas exchange system and will be briefly included here.

7.1. Odor acquisition

Humans show a remarkable ability to "tune out" odors (Zelano et al., 2005) despite the fact that they have excellent olfactory sensitivity (Lorig, 1999). Recent evidence from several different laboratories has demonstrated odor-related changes in brain activity despite an absence of awareness for the odor (Kirk-Smith et al., 1983; Lorig et al., 1990, 1991; Sobel et al., 1999; Kline et al., 2000; Jacob et al., 2001). With clear evidence of odor effects even when subjects don't notice their presence, it appears likely that each inhalation brings new olfactory information to the nervous system. Certainly, this phenomenon is true of other animals. Freeman and Schnieder (1982) and also Chaput (1986) found that the pattern of electrical activity on the olfactory bulb of rabbits was dependent on respiratory cycle. They suggested that the initiation of an inspiration triggered a "template" related to the odor to which the animal was conditioned. Buonviso et al. (2003) found that the primary component of rhythmic activity in the bulb of rats was dependent upon the respiratory cycle. Lorig and colleagues (Lorig et al., 1996), examined the effect of breathing pattern on chemosensory evoked potentials in humans and found that they differed as a function of whether stimuli were delivered in concert with inspiration or out of phase with this cycle further suggesting the production of such an odor template. In a related fMRI study, Sobel and coworkers (Sobel et al., 1998) found that sniffing produced activation patterns different than smelling, further illustrating that respiratory activity serves to synchronize a variety of neural sub-systems related to odors and their anticipation.

In a recent study, Kay (2005) describes findings that illustrate how the olfactory-respiratory system is controlled. Based primarily on coherence data, it appears that as odor becomes available, activity in the bulb triggers hippocampal activity that forms a re-entrant loop with the bulb and also descends to influence control of the diaphragm. The odor signals an increase in respiratory effort resulting in sniffing. In a curious way, this process is similar to vision. Availability of visual input leads to movement of the eyes to focus the image on the fovea leading to better discrimination. In the case of olfactory stimuli, sniffing is the movement and this changes the dynamics of airflow in the nasal cavity leading to a change in the distribution of chemicals in the mucosa. Although we have not yet found the olfactory epithelium analog of the fovea, the behavior equivalent is clearly present because previous behavioral research has indicated that sniffing is the most efficient and information-rich way in which to evaluate odors (Laing, 1983).

Recently, it has been suggested that respiratory activity is reflexively truncated in response to a bad odor (Frank et al., 2003). For some time, it has been known that airborne malodors and irritants can affect breathing including reducing tidal volume (Warren et al., 1992) but only recently has a reflexive mechanism been proposed that might be related to odor hedonics. The argument for the reflexive nature is based on the rapidity of this inspiratory cessation. Inhalatory restriction starts at approximately 300 msec following odor availability. Reaction times to these odors tend to be more than 100 msec later (Overbosch et al., 1989) suggesting that the cessation takes place well before detection. Although this is hardly conclusive evidence of a reflexive connection, it does raise the strong possibility that some hedonic information is encoded prior to the action of central nervous system resources necessary for odor detection.

7.2. Speech

Another important aspect of breathing is providing a medium for speech sounds. Although often overlooked as a function of breathing, this is an important problem for subjects requiring artificial ventilation. Coordination of laryngeal movements with enforced lung movements and the availability of an air supply is often difficult to learn. Relatively few reports describe the pathways and mechanisms by which ventilatory movements are coordinated with speech production. It seems likely that such coordination is a product of feedback between cortical speech areas and the areas associated with voluntary breathing although a recent study with cats highlights the importance of the midbrain. Davis and colleagues (Davis et al., 1996) found that the periaqueductal gray area was associated to both respiratory and laryngeal output and their findings suggest that it integrates activity for these sets of muscles.

8. CONCLUSIONS

Like the actions of the heart, respiration is clearly a critical and life sustaining process. As might be expected of any process so important to survival, it is multiply controlled and affects many systems of the body. Likewise, it receives broad input for its regulation. Traditionally, psychophysiology has concentrated on other dependent variables. When respiration was measured it was often used as a covariate or control for the variables of primary interest. Clinical medicine, however, routinely examines the activity of this system and has developed many effective strategies for its assessment. It is within medicine that most of the advances in respiratory measurement have been made in the past few decades.

It is always difficult to imagine what the future holds in a research area. Technical and computational advances will certainly continue. There are already strong trends for more portable devices to measure respiration. Additionally, it is likely that the availability of high level mathematical programming languages designed to work with real data (e.g., MATLAB, LabView, etc.) will have a major impact on the ways that this seemingly simple waveform are viewed. It is also clear that advances such as ambulatory monitoring will lead to new and important insights about respiration in the real world. The fact that a system, ambulatory or not, provides simultaneous measures of so many interrelated dependent variables will beg for more sophisticated analyses than are currently being conducted. One might imagine a mathematical function indicating a parameter called metabolic respiratory demand. Such a function would be based on the long-term, nonlinear multiple regression between blood oxygen saturation, cardiac output and inspiratory volume. Knowing two of the variables should predict the third and the deviance between the predicted and actual values for inspiratory volume might be an indicator of "voluntary override." It would certainly be a curious twist of fate that the characteristic that turned early psychophysiologists away from studying this system (Wientjes & Grossman, 1998), would become an important area of study in the following century.

It is genuinely puzzling that a system with such broad effects on the brain and body is so relatively unstudied by psychophysiologists. There is far more to this system than it appears. Perhaps, as Wientjes and Grossman (1998) have suggested, we will find new insights into the areas of emotion and cognition in the loose coupling of respiration to the other systems of our bodies.

REFERENCES

Allen, K. (2004). Principles and limitations of pulse oximetry in patient monitoring. *Nursing Times, 100*, 34–37.

Aloia, M. S., Arnedt, J. T., Davis, J. D., Riggs, R. L., & Byrd, D. (2004). Neuropsychological sequelae of obstructive sleep apnea-hypopnea syndrome: a critical review. *Journal International Neuropsychological Society, 10*, 772–785.

Bender, B. G., & Annett, R. D. (1999). Neuropsychological outcomes of nocturnal asthma. *Chronobiology International, 16*, 695–710.

Berntson, G. G., Cacioppo, J. T., & Quigley, K. S. (1993). Respiratory sinus arrythmia: Autonomic origins, psychological mechanisms, and psychophysiological implications. *Psychophysiology, 30*, 183–196.

Berntson, G. G., & Stowell, J. R. (1998). ECG artifacts and heart period variability: don't miss a beat. *Psychophysiology, 35*, 127–132.

Block, R. A., Arnott, D. P., Quigley, B., & Lynch, W. C. (1989). Unilateral nostril breathing influences lateralized cognitive performance. *Brain and Cognition, 9*, 181–190.

Boiten, F. A,. Frijda, N. H., & Wientjes, C. J. (1994). Emotions and respiratory patterns: review and critical analysis. *International Journal of Psychophysiology, 17*, 103–128.

Bramann, S. S. (1995). The regulation of normal lung function. *Allergy Procedings*. *16*, 223–226.

Brosch, J. R., Talavage, T. M., Ulmer, J. L., & Nyenhuis, J. A. (2002). Simulation of human respiration in fMRI with a mechanical model. *IEEE Transactions on Biomedical Engineering, 49*, 700–707.

Buonviso, N., Amat, C., Litaudon, P., Roux, S., Royet, J. P., Farget, V., & Sicard, G. (2003). Rhythm sequence through the olfactory bulb layers during the time window of a respiratory cycle. *European Journal of Neuroscience, 17*, 1811–1819.

Carroll, N., Clague, J. E., Pollard, M. N., Horan, M. A., Edwards, R. H., & Calverley, P. M. (1992). Portable maximum respiratory pressure measurement – a comparison with laboratory techniques. *Journal of Medical Engineering Technology, 16*, 82–86.

Casili, J. G., Wierwille, W. W., & Cordes, R. E. (1983). Respiratory measurement: Overview and new instrumentation. *Behavior Research Methods, Instruments, and Computers, 15*, 401–405.

Chaput, M. A. (1986). Respiratory-phase-related coding of olfactory information in the olfactory bulb of awake freely-breathing rabbits. *Physiology and Behavior, 36*, 319–324.

Collet, C., Deschaumes-Molinaro, C., Delhomme, G., Dittmar, A., & Vernet-Maury, E. (1999). Autonomic responses correlate to motor anticipation. *Behavioral Brain Research, 29*(63), 71–79.

Comroe, J. H. (1974). *The physiology of respiration, 2nd edition*. Chicago: Year Book Medical Publishers.

Davis, C., Mazzolini, A., Mills, J., & Dargaville, P. (1999). A new sensor for monitoring chest wall motion during high-frequency oscillatory ventilation. *Medical Engineering Physics, 21*, 619–623.

Davis, P. J., Zhang, S. P., Winkworth, A., & Bandler, R. (1996). Neural control of vocalization: respiratory and emotional influences. *Journal of Voice, 10*, 23–38.

De Troyer, A., Gorman, R. B., & Gandevia, S. C. (2003). Distribution of inspiratory drive to the external intercostals muscles in humans. *Journal of Physiology, 546*, 943–954.

De Troyer, A., Kelly, S., Macklem, P. T., Zin, W. A. (1985). Mechanics of intercostal space and actions of external and internal intercostal muscles. *Journal of Clinical Investigation, 75*, 850–7.

Dudley, D. L. (1969). *Psychophysiology of respiration in health and disease*. New York: Appelton-Century-Crofts.

Eccles, R. (1996). A role for the nasal cycle in respiratory defence. *European Respiration Journal, 9*, 371–379.

Ernst, J. M., Litvack, D. A., Lozano, D. L., Cacioppo, J. T., & Berntson, G. G. (1999). Impedance pneumography: Noise as signal in impedance cardiography. *Psychopysiology, 36*, 333–338.

Evans, S. E., & Scanlon, P. D. (2003). Current practice in pulmonary function testing. *Mayo Clinic Procedings,* 78, 758–763.

Feleky, A. (1916). The influence of emotions on respiration. *Journal of Experimental Psychology, 1*, 218–222.

Frank, R. A., Dulay, M. F., & Gestland, R. C. (2003). Assessement of the sniff magnitude test as a clinical test of olfactory function. *Physiology and Behavior, 78*, 195–204.

Freeman, W. J., & Schneider, W. (1982). Changes in spatial patterns of rabbit olfactory EEG with conditioning to odors. *Psychophysiology, 19*, 44–56.

Fung, M.-L., & St. John, W. M. (1995). Expiratory neural activities in gasping induced by pharyngeal stimulation and hypoxia. *Respiration Physiology, 100,* 119–127.

Gomez, P., Stahel, W. A., Danuser, B. (2004). Respiratory responses during affective picture viewing. *Biological Psychology, 67*, 359–373.

Grossman, P. (1983). Respiration, stress and cardiovascular function. *Psychophysiology, 20*, 284–300.

Grossman, S. A. (1967). *A textbook of physiological psychology*. New York: Wiley.

Guyton, A. C., & Hall, J. E. (2000). *Textbook of Medical Physiology, 10th ed*, New York: W. B. Saunders.

Han, J. N., Stegen, K., Cauberghs, M., & Vande Woestijne, K. P. (1997). Influence of awareness of the recording of breathing on respiratory pattern in healthy humans. *European Respiratory Journal,* 10, 161–166.

Harada, Y., Kuno, M., & Wang, Y. Z. (1995). Differential effects of carbon dioxide and pH on central chemoreceptors in the rat in vitro. *Journal of Physiology, 368*, 679–693.

Harver, A., & Lorig, T. S. (2000). Respiration. In Cacioppo, et al. (eds.), *The Handbook of Psychophysiology, 2nd ed.*, pp. 265–293.

Harver, A., & Mahler, D. A. (1998). Perception of increased resistance to breathing. In H. Kostes & A. Harver (eds.), *Self-management of asthma*. New York: Marcel Dekker, pp. 147–193.

Hlastala, M. P., & Berger, A. J. (1996). *Physiology of respiration*. New York: Oxford University Press.

Jella, S. A., & Shannahoff-Khalsa, D. S. (1993). The effects of unilateral forced nostril breathing on cognitive performance. *International Journal of Neuroscience, 73*, 61–68.

Jacob, S., Kinnunen, L. H., Metz, J., Cooper, M., & McClintock, M. K. (2001). Sustained human chemosignal unconsciously alters brain function. *Neuroreport, 8*, 2391–2394.

Johnson, R. L. Jr., & Miller, J. M. (1968). Distribution of ventilation, blood flow and gas transfer coefficient. *Journal of Applied Physiology, 25*, 1–15.

Kay, L. M. (2005). Theta oscillations and sensorimotor performance. *Procedings of the National Academies of Science, 102*, 3863–3868.

Kirk-Smith, M., Van Toller, C., & Dodd, G. (1983). Unconscious odour conditioning in human subjects. *Biological Psychology, 17*, 221–231.

Kline, J. P., Schwartz, G. E., Dikman, Z. V., & Bell, I. R. (2000). Electroencephalographic registration of low concentrations of isoamyl acetate. *Consciousness and Cognition, 9*, 50–65.

Laing, D. G. (1983). Natural sniffing gives optimum odour perception for humans. *Perception, 12*, 99–117,

Lehrer, P., Feldman, J., Giardino, N., Song, H. S., & Schmaling, K. (2002). Psychological aspects of asthma. *Journal of Consulting and Clinical Psychology, 70*, 691–711.

Levine, S., Silage, D., Henson, D., Wang, J. Y., Krieg, J., LaManca, J., & Levy, S. (1991). Use of a triaxial magnetometer for respiratory measurements. *Journal of Applied Physiology, 70*, 2311–2321.

Lorig, T. S. (1999). On the similarity of odor and language processing. *Neuroscience and Biobehavioral Reviews, 23*, 391–398.

Lorig, T. S., Herman, K. B., Schwartz, G. E., & Cain, W. S. (1990). EEG activity during administration of low concentration odors. *Bulletin of the Psychonomic Society, 28*, 405–408.

Lorig, T. S., Huffman, E., DeMartino, A., & DeMarco, J. (1991). The effects of low concentration odors on EEG activity and behavior. *Journal of Psychophysiology, 5*, 69–77.

Lorig, T. S., Malin, E. L., & Horwitz, J. E. (2005). Odor mixture alters neural resources during symbolic problem solving. *Biological Psychology, 69,* 205–216.

Lorig, T. S., Matia, D. C., Peszka, J. J., & Bryant, D. N. (1996). The effects of active and passive stimulation on chemosensory event-related potentials. *International Journal of Psychophysiology, 23*, 199–205.

Lorig, T. S., Schwartz, G. E., Herman, K. B., & Lane, R. (1988). Brain and Odor II: EEG activity during nose and mouth breathing. *Psychobiology, 16*, 285–287.

McCool, D. F., Wang, J., & Ebi, K. L. (2002). Tidal volume and respiration timing derived from a portable ventilation monitor. *Chest, 122*, 684–691.

Mortimer, K. M., Fallot, A., Balmes, J. R., & Tager, I. B. (2003). Evaluating the use of a portable spirometer in a study of pediatric asthma. *Chest, 123*, 1899–1907.

Nielsen, J., & Roth, P. (1929). Clinical spirography. *Archive of Clinical Medicine, 43*, 132–138.

Overbosch, P., de Wijk. R., de Jonge, T. J., & Koster, E. P. (1989). Temporal integration and reaction times in human smell. *Physiology and Behavior, 45*, 615–626.

Pauly, J. E. (1957). Electromyographic studies of human respiration, *Chicago Medical School Quarterly, 18*, 80–88.

Pennock, B. E. (1990). Rib and abdominal piezoelectric film belts to measure ventilatory airflow. *Journal of Clinical Monitoring, 6*, 276–283.

Que, C. L., Kolmaga, C., Durand, L. G., Kelly, S. M., & Macklem, P. T. (2002). Phonospirometry for noninvasive measurement of ventilation: methodology and preliminary results. *Journal of Applied Physiology, 93*, 1515–1526.

Richards, D. W., Jr. (1953). The nature of cardiac and pulmonary dsypnea. *Circulation, 7*, 15–29.

Ries, A. L. (1989). Measurement of lung volumes. *Clinical Chest Medicine, 10*, 177–186.

Rittweger, J., Lambertz, M., & Langhorst, P. (1997). Influences of mandatory breathing on rhythmical components of electrodermal activity. *Clinical Physiology,* 17, 609–618.

Ritz, Thomas (2004). Probing the psychophysiology of the airways: Physical activity, experience emotion and facially expressed emotion. *Psychophysiology, 41*, 809–821.

Ritz, T., Dahme, B., Dubois, A. B., Folgering, H., Fritz, G. K., Harver, A., Kotses, H., Lehrer, P. M., Ring, C., Steptoe, A., & Van de Woestijne, K. P. (2002). Guidelines for mechanical lung function measurements in psychophysiology. *Psychophysiology, 39*, 546–67.

Rosanski, E. A., & Hoffman, A. M. (1999). Pulmonary function testing in small animals. *Clinical Technique in Small Animal Practice, 14*, 237–41.

Saucier, D. M., Tessem, F. K., Sheerin, A. H., & Elias, L. (2004). Unilateral forced nostril breathing affects dichotic listening for emotional tones. *Brain and Cognition, 55*, 403–405.

Sasaki, C. T., & Mann, D. G. (1976). Dilator naris function. *Archive of Otolaryngology, 102*, 365–367.

Schneider, R., Schmidt, S., Binder, M., Schafer, F., & Walach, H. (2003). Respiration-related artifacts in EDA recordings: introducing a standardized method to overcome multiple interpretations. *Psychological Reports, 93*, 907–920.

Sinha, R., Lovallo, W. R., & Parsons, O. A. (1992). Cardiovascular differentiation of emotions. *Psychosomatic Medicine, 54*, 422–435.

Slosman, D. O., De Ribaupierre, S., Chicherio, C., Ludwig, C., Montandon, M. L., Allaoua, M., Genton, L., Pichard, C., Grousset, A., Mayer, E., Annoni, J. M., & De Ribaupierre, A. (2004). Negative neurofunctional effects of frequency, depth and environment in recreational scuba diving: the Geneva "memory dive" study. *British Journal of Sports Medicine, 38*, 108–114.

Sobel, N., Prabhakaran, V., Desmond, J. E., Glover, G. H., Goode, R. L., Sullivan, E. V., & Gabrieli, J. D. (1998). Sniffing and smelling: Separate subsystems in human olfactory cortex. *Nature, 392*, 282–286.

Sobel, N., Prabhakaran, V., Hartley, C. A., Desmond, J. E., Glover, G. H., Sullivan, E. V., & Gabrieli, J. D. (1999). Blind smell: brain activation induced by an undetected air-borne chemical. *Brain, 122*, 209–217.

Sternbach, G. L., Varon, J., Fromm, R. E., Sicuro, M., & Basket, P. J. (2001). Galen and the origins of artificial ventilation, the arteries and the pulse. *Resuscitation, 49*, 119–122.

Sutherland, G. F., Wolf, A., & Kennedy, F. (1938). The respiratory fingerprint of the nervous system. *Medical Record, 148*, 101–103.

Shea, S. A., & Guz, A. (1992). *Personnalite ventilatoire* – An overview. *Respiration Physiology, 87*, 275–291.

Warren, D. W., Walker, J. C., Drake, A. F., & Lutz, R. W. (1992). Assessing the effects of odorants on nasal airway size and breathing. *Physiology and Behavior, 51*, 425–430.

Werntz, D. A., Bickford, R. G., & Shannahoff-Khalsa, D. (1987). Selective hemispheric stimulation by unilateral forced nostril breathing. *Human Neurobiology, 6*, 165–171.

Wientjes, C. J. F., & Grossman, P. (1998). Respiratory psychophysiology as a discipline: introduction to the special issue. *Biological Psychology, 49*, 1–8.

Wilhelm, F. H., Grossman, P., & Coyle, M. A. (2004). Improving estimation of cardiac vagal tone during spontaneous breathing using a paced breathing calibration. *Biomedical Science Instrumentation 40*, 317–24.

Wilhelm, F. H., Roth, W. T., & Sackner, M. A. (2003). The LifeShirt. An advanced system for ambulatory measurement of respiratory and cardiac function. *Behavior Modification, 23*, 671–679.

Wilson, T. A., Legrand, A., Gevenois, P. A., & De Troyer, A. (2001). Respiratory effects of the external and internal intercostal muscles in humans. *Journal of Physiology, 530*, 319–330.

Zelano, C. Bensafi, M., Porter, J., Mainland, J., Johnson, B., Bremner, E., Telles, C., Khan, R., & Sobel, N. (2005). Attentional modulation in human primary olfactory cortex. *Nature Neuroscience, 8*, 114–120.

Zhu, H., Goodyear, B. G., Lauzon, M. L., Brown, R. A., Mayer, G. S., Law, A. G., Mansinha, L., & Mitchell, J. R. (2003). *Medical Physics, 30*, 1134–1141.